Linux Fundamentals

About Linux

What is Linux?

Linux is **a free, open-source operating system (OS) based on Unix.**

What makes Linux uniquely powerful and distinct from proprietary systems like Microsoft Windows or macOS include the following traits:

- **Fully Open-Source**
- Multi-Platform
- **Highly Customizable & Lightweight**
- Secure and stable
- Community-Driven
- Kernel-Based

Why Learn Linux?

- Industry Standard
- Career Opportunity
- Security and Stability
- Customizable
- Efficient

List of Linux Distributions

- Debian-Based
  Ubuntu
  Kali-Linux (for pentesting)
  Linux Mint (Ubuntu-based)
  Parrot OS
  Kali Purple (for both Blue & Red)
- Red Hat Enterprise Linux
  Fedora Linux

Rocky Linux
Alma Linux
CentOS

- Arch Linux
  Manjaro
- openSUSE
  SUSE Linux Enterprise
- Gentoo
- Slackware

Lab 01 - Linux Navigation

Commands

Command	Definition	Example
`pwd`	Prints the current working directory.	`pwd`
`ls`	Lists files and directories.	`ls`
`ls -l`	Lists files in long format with permissions and ownership.	`ls -l`
`ls -a`	Shows hidden files and directories.	`ls -a`
`cd`	Changes the current directory.	`cd Documents`
`cd ..`	Moves one directory up.	`cd ..`
`cd ~`	Moves to the user's home directory.	`cd ~`
`cd /`	Moves to the root directory.	`cd /`
`tree`	Displays the directory structure as a tree.	`tree`

Command	Definition	Example
`clear`	Clears the terminal screen.	`clear`
`history`	Shows previously executed commands.	`history`

Absolute Path vs Relative Path

cd /home/Documents	Absolute Path
cd Documents	Relative Path

Practical

```
pwd
ls
cd ~
pwd
cd /
pwd
cd ~
ls -a
history
clear
```

Lab 02 – File Management

This is where you learn how to create, copy, move, view, and delete files.

Commands Covered

Command	Definition	Example
`touch`	Creates an empty file.	`touch notes.txt`

Command	Definition	Example
`mkdir`	Creates a directory.	`mkdir projects`
`rmdir`	Removes an empty directory.	`rmdir projects`
`cp`	Copies files or directories.	`cp notes.txt backup.txt`
`mv`	Moves or renames files.	`mv notes.txt linux_notes.txt`
`rm`	Removes files.	`rm notes.txt`
`rm -r`	Removes directories recursively.	`rm -r testfolder`
`cat`	Displays file contents.	`cat notes.txt`
`less`	Views file contents page by page.	`less notes.txt`
`head`	Shows first 10 lines of a file.	`head notes.txt`
`tail`	Shows last 10 lines of a file.	`tail notes.txt`
`echo`	Outputs text or writes to files.	`echo "Hello" > notes.txt`
`file`	Identifies file type.	`file notes.txt`

Practical

```
mkdir linux-lab
cd linux-lab
touch notes.txt
echo "Linux Fundamentals" > notes.txt
cat notes.txt
cp notes.txt backup.txt
mv backup.txt notes_backup.txt
ls
head notes.txt
tail notes.txt
file notes.txt
rm notes_backup.txt
```

```
cd ..
rm -r linux-lab
```

Lab 03 – Users & Groups

permissions and access control are built around users and groups.

Commands Covered

Command	Definition	Example
`whoami`	Shows the current logged-in user.	`whoami`
`id`	Shows user ID (UID) and group ID (GID).	`id`
`groups`	Displays groups a user belongs to.	`groups`
`sudo`	Runs a command with elevated privileges.	`sudo apt update`
`useradd`	Creates a new user.	`sudo useradd user`
`passwd`	Sets or changes a user's password.	`sudo passwd user`
`usermod`	Modifies a user account.	`sudo usermod -aG sudo user`
`groupadd`	Creates a new group.	`sudo groupadd developers`
`groupdel`	Deletes a group.	`sudo groupdel developers`
`userdel`	Deletes a user.	`sudo userdel user`
`su`	Switches to another user.	`su user`

Practical

```
whoami
id
```

```
groups
sudo groupadd cyber
sudo useradd k-lab
sudo usermod -aG cyber k-lab
id k-lab
su k-lab
whoami
```

Lab 04 – File Permissions

Many privilege escalation vulnerabilities happen because of misconfigured permissions.

Commands Covered

Command	Definition	Example
`ls -l`	View file permissions.	`ls -l file.txt`
`chmod`	Change file permissions.	`chmod 755 script.sh`
`chown`	Change file owner.	`sudo chown user file.txt`
`chgrp`	Change file group.	`sudo chgrp developers file.txt`
`umask`	View/set default permissions.	`umask`

```
-rw-r--r--
| |  |  |
| |  |  └── Others
| |  └───── Group
| └──────── Owner
└────────── File Type
```

Number	Permission
777	rwxrwxrwx
755	rwxr-xr-x
700	rwx------
644	rw-r--r--
600	rw-------

Practical

```
mkdir permission-lab
cd permission-lab
touch file1.txt
touch file2.txt
ls -l
chmod 644 file1.txt
chmod 777 file2.txt
ls -l
touch script.sh
chmod +x script.sh
ls -l
sudo chown root file1.txt
ls -l
sudo chgrp cyber file1.txt
ls -l
umask
```

Lab 05 – Networking Basics

Commands Covered

Command	Definition	Example
`ip a`	Displays network interfaces and IP addresses.	`ip a`
`ip route`	Shows routing table.	`ip route`
`hostname`	Displays system hostname.	`hostname`
`ping`	Tests connectivity to a host.	`ping google.com`
`ss`	Shows network connections and listening ports.	`ss -tuln`
`netstat`	Displays network statistics and ports.	`netstat -tuln`
`nslookup`	Queries DNS records.	`nslookup google.com`
`dig`	Advanced DNS lookup tool.	`dig google.com`
`curl`	Transfers data from URLs.	`curl https://google.com`
`wget`	Downloads files from the web.	`wget https://google.com/file.txt`

Practical

```
ip a
ip route
hostname
ping google.com
ss -tuln
nslookup google.com
dig google.com
curl https://google.com
wget https://google.com
```

t = TCP

u = UDP

l = Listening

n = Numeric

Lab 06 - Processes & Services

This lab teaches you how Linux runs programs and manages services. In cybersecurity, you'll often need to find, monitor, or stop processes.

Commands Covered

Command	Definition	Example
`ps`	Displays running processes.	`ps`
`ps aux`	Shows all running processes.	`ps aux`
`top`	Real-time process monitor.	`top`
`htop`	Improved interactive process viewer.	`htop`
`pgrep`	Find process IDs by name.	`pgrep firefox`
`kill`	Terminates a process.	`kill 1234`
`kill -9`	Forcefully kills a process.	`kill -9 1234`
`jobs`	Lists background jobs.	`jobs`
`bg`	Sends a job to background.	`bg %1`
`fg`	Brings a job to foreground.	`fg %1`
`systemctl`	Manages system services.	`systemctl status ssh`
`service`	Legacy service management.	`service ssh status`

Practical

```
ps
ps aux
top (q for exit)
```

```
htop (F10 for exit)
sleep 300
kill 1234
ps aux | grep sleep
sleep 300 &
jobs
fg &
systemctl status ssh
```

Lab 07 - Package Management

Package management is how Linux installs, updates, and removes software.

Commands Covered

Command	Definition	Example
`apt update`	Updates package information.	`sudo apt update`
`apt upgrade`	Upgrades installed packages.	`sudo apt upgrade`
`apt install`	Installs a package.	`sudo apt install tree`
`apt remove`	Removes a package.	`sudo apt remove tree`
`apt purge`	Removes package and configuration files.	`sudo apt purge tree`
`apt autoremove`	Removes unused dependencies.	`sudo apt autoremove`
`apt search`	Searches for packages.	`apt search nmap`
`apt show`	Displays package information.	`apt show nmap`
`dpkg -l`	Lists installed packages.	`dpkg -l`

Lab 08 - Log Analysis

Logs are one of the most important sources of information for troubleshooting, incident response, and security investigations.

Commands Covered

Command	Definition	Example
`journalctl`	View system logs managed by systemd.	`journalctl`
`journalctl -xe`	View recent system events and errors.	`journalctl -xe`
`dmesg`	View kernel messages.	`dmesg`
`tail`	View the end of a file.	`tail auth.log`
`tail -f`	Follow log updates in real time.	`tail -f auth.log`
`grep`	Search for text patterns.	`grep ssh auth.log`
`less`	Read large log files page by page.	`less auth.log`
`cat`	Display entire file contents.	`cat auth.log`

File	Purpose
`/var/log/syslog`	General system logs
`/var/log/auth.log`	Authentication logs
`/var/log/kern.log`	Kernel logs
`/var/log/dpkg.log`	Package installation logs
`/var/log/boot.log`	Boot logs

Lab 9 – SSH (Secure Shell)

SSH allows you to securely access another Linux machine remotely.

Commands Covered

Command	Definition	Example
`ssh`	Connect to a remote machine.	`ssh user@192.168.1.10`
`ssh-keygen`	Generate SSH key pair.	`ssh-keygen`
`ssh-copy-id`	Copy public key to remote machine.	`ssh-copy-id user@192.168.1.10`
`scp`	Securely copy files.	`scp file.txt user@host:/home/user/`
`sftp`	Secure file transfer session.	`sftp user@host`

Lab 10 – Bash Scripting Basics

Commands & Concepts Covered

Concept	Definition	Example
Variable	Store data in memory	`name="Kartik"`
echo	Display output	`echo $name`
read	Take user input	`read name`
if	Conditional statement	`if [ $a -eq $b ]`
for	Loop through items	`for i in {1..5}`
while	Run while condition is true	`while [ $x -lt 5 ]`
function	Reusable block of code	`greet(){}`
chmod +x	Make script executable	`chmod +x script.sh`
./	Execute script	`./script.sh`

```
nano hello.sh
```

```
#!/bin/bash

echo "Hello, World!"
```

```
chmod +x hello.sh
./hello.sh
```

```
#!/bin/bash

name="User"

echo "Hello $name"
```

```
#!/bin/bash

echo "Enter your name:"
read name

echo "Welcome $name"
```

```
#!/bin/bash

num=10

if [ $num -eq 10 ]
then
    echo "Number is 10"
fi
```

```
#!/bin/bash

for i in {1..5}
do
    echo $i
done
```

```bash
#!/bin/bash

count=1

while [ $count -le 5 ]
do
    echo $count
    count=$((count+1))
done
```

```bash
#!/bin/bash

greet() {
    echo "Welcome to Linux"
}

greet
```

```
Project:
nano system-info.sh
```

```bash
#!/bin/bash

echo "===== SYSTEM INFO ====="

echo "Hostname:"
hostname

echo ""

echo "Current User:"
whoami
```

```
echo ""

echo "IP Address:"
hostname -I
```

```
chmod +x system-info.sh
./system-info.sh
```

```
#!/bin/bash

echo "Current User: $(whoami)"
echo "Home Directory: $HOME"
echo "Current Directory: $(pwd)"
echo "Date: $(date)"
```

Help commands

- <tool> —help
- <tool> -h
- apropos <tool>

Note: Some Command need sudo if permission denied.